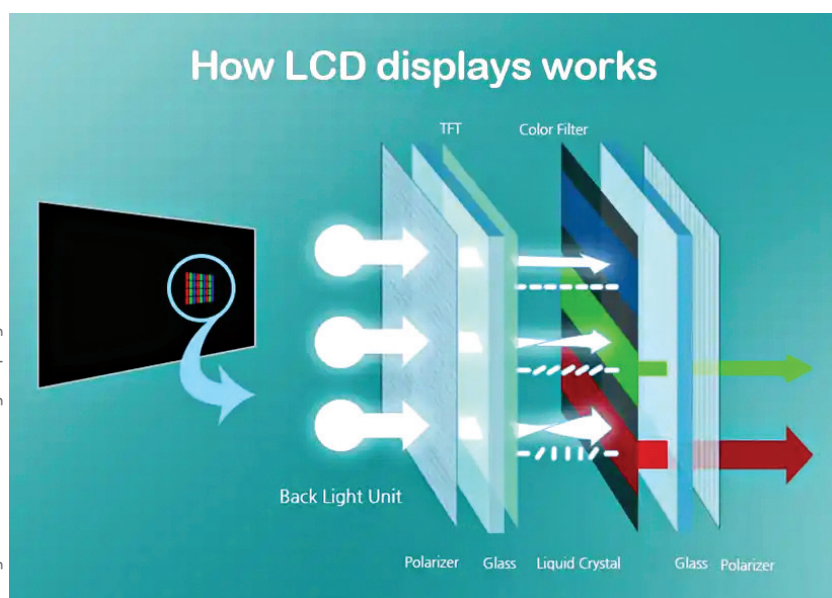


Image Source: Samsung Displays



model. Another notable difference is that LG uses IPS LCD panels and Samsung's QLED displays can have both IPS or VA LCD panels. VA panels offer higher contrast but poorer viewing angles, while IPS LCDs offer wider viewing angles but lower contrast.

Sony's Triluminous series of displays debuted in 2013, Sony used Quantum Dots to enhance LED backlight, but the current Triluminous Pro, XR Triluminous Pro and XR Triluminous Max displays don't seem to use QDEF.

There is another QNED

The term QNED is embroiled in a bit of controversy. Before LG registered the QNED Trademark for its displays, QNED was popular as Quantum Nanorod Emitting Diode, a self-emissive Samsung Display technology that is expected to succeed their latest Quantum-Dot OLED or QD-OLED displays.

QD-OLED displays work by using a blue light-emitting layer made of OLEDs. Then, they add a layer of printed Quantum Dots to create pure red and green sub-pixels. This is different from QLEDs, where the Quantum Dots layer is placed before the backlight reaches the LCD substrate. In QD-OLEDs, the Quantum Dots layer comes after the subpixels, facing the user.

The next Generation QNED displays being developed by Samsung are expected to replace the Blue OLED emission later with tiny Inorganic Blue GaN LEDs or nanorods that will be more durable and enable higher brightness. These LEDs are like tiny towers made of GaN that can emit light more efficiently.

FAQs

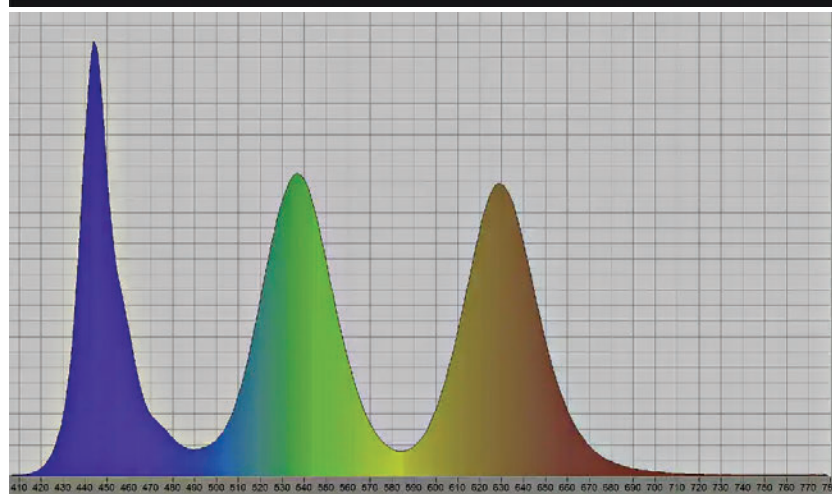
Q) Is QNED better than QLED?

QNED and QLED are similar technologies. Which one is better will depend on the specific models that you are comparing.

Q) Is OLED better than QLED or QNED?

Simply put, yes. Most modern OLED TVs offer a better TV viewing experience and impactful HDR as compared to LED LCD TVs. 

Source: HDTVTest



Distinct Red, Green and Blue Waveforms in Samsung QLED TV

Display Type	QNED	QLED
Quantum Dot Enhancement Film	Yes	Yes
Nanocell technology	Yes	No
Backlight	Mini LED, Edge-lit LED, Full Array LED	Mini LED, Edge-lit LED, Full Array LED
LCD Type	IPS LCD	VA LCD (mostly), IPS LCD
Brands	LG	Samsung, Hisense, TCL & others